

DEPARTMENT OF ELECTRONICS AGH

L A B O R A T O R Y
ELECTRONIC DEVICES

MOS FIELD-EFFECT
TRANSISTOR

REV. 2.4

1 Aim of the Exercise

- Getting acquainted with the operation of the MOS field-effect transistor.
- Plotting the current–voltage characteristics of the n-MOSFET and p-MOSFET transistors that form complementary pairs in the CD4007 integrated circuit (Fig. 1).
- Determination of the basic parameters of these transistors from their characteristics.

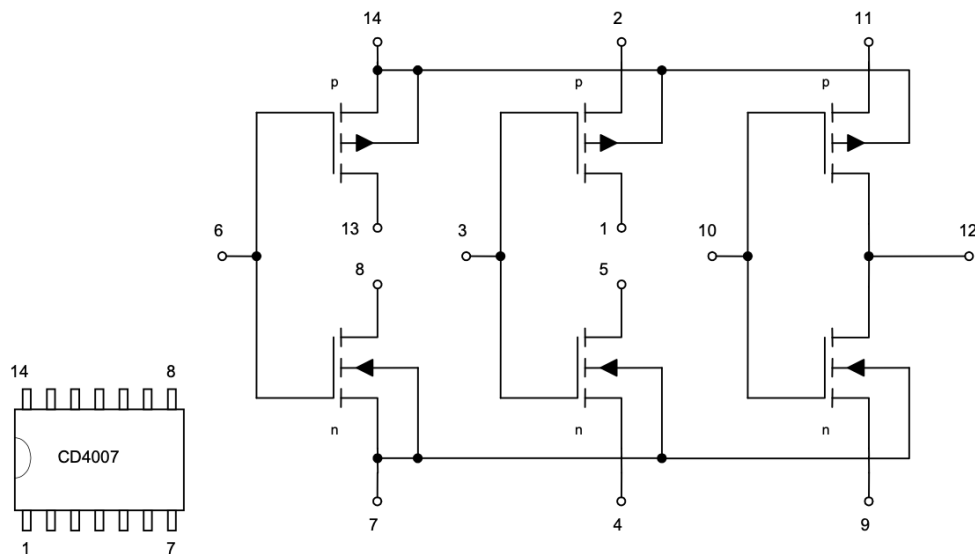


Figure 1: Internal schematic of the CD4007 integrated circuit.

2 Equipment and Components

The following instruments and components will be used during the exercise:

- NI ELVIS Prototyping Board connected to a PC,
- I-V Plotter (dedicated measurement application — IV3 available on the computers),
- external DC power supply,
- components listed in Table 1.

Table 1: Component values for the exercise.

Component	Value / Type
Integrated circuit	CD4007 (or equivalent)
Resistor	100 kΩ

3 Preparation

- 3.1. Draw the transfer and output characteristics of an enhancement-mode n-channel and p-channel MOS transistor.
- 3.2. Using the NI ELVIS breadboard layout, prepare assembly diagrams for the measurement circuits used in this exercise.

4 Laboratory Procedure

4.1 Measurement of the Transfer Characteristics of the n-MOS Transistor

Assemble the measurement circuit according to Fig. 2, selecting one of the transistors available in the CD4007 integrated circuit. The resistor R is used to protect the transistor against damage; its recommended value is $R = 100\text{ k}\Omega$.

Using the I-V Plotter, measure a family of transfer characteristics of the n-MOSFET for positive gate voltages U_{GS} ranging from 0 to 10 V, for several values of U_{DS} in the range 0 to 5 V. The U_{GS} step should be approximately 0.2 V (swept automatically by the I-V Plotter), while U_{DS} should be changed in steps of 2 V (set manually on the external power supply), or as instructed by the supervisor.

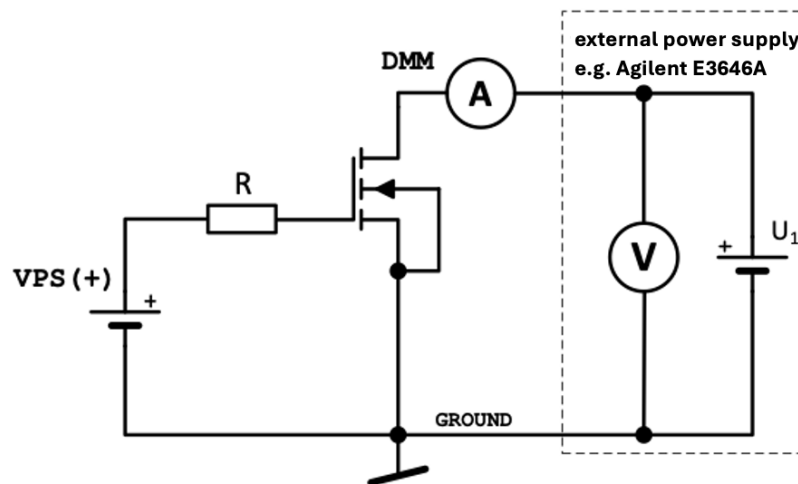


Figure 2: Measurement circuit for the transfer characteristic of the n-MOS transistor.

4.2 Measurement of the Output Characteristics of the n-MOS Transistor

Assemble the measurement circuit according to Fig. 3, selecting one of the transistors available in the CD4007 integrated circuit. Using the I-V Plotter, measure a family of output characteristics of the n-MOSFET for U_{DS} ranging from 0 to 10 V, for several fixed gate voltages (e.g. $U_{GS} = 2, 4, 6, 8, 10\text{ V}$, step $\Delta U_{DS} = 0.2\text{ V}$), or as instructed by the supervisor (do not exceed $U_{DS\text{ max}} = 15\text{ V}$).

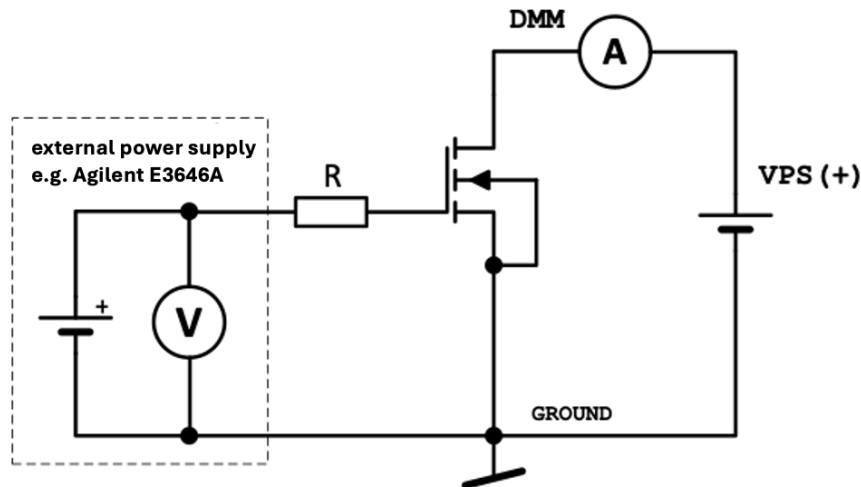


Figure 3: Measurement circuit for the output characteristics of the n-MOS transistor.

4.3 Investigation of the Body Effect in the n-MOS Transistor

The transistor under test should operate in the saturation region. These conditions are ensured by the circuit shown in Fig. 4. Assemble the measurement circuit according to Fig. 4, selecting one of the transistors available in the CD4007 integrated circuit.

Using the I-V Plotter, measure a family of transfer characteristics of the n-MOSFET for positive gate voltages U_{GS} ranging from 0 to 10 V, for several fixed values of the bulk-source voltage U_{BS} (e.g. $U_{BS} = 0, -2, -4, -6, -8, -10$ V, step $\Delta U_{GS} = 0.2$ V), or as instructed by the supervisor.

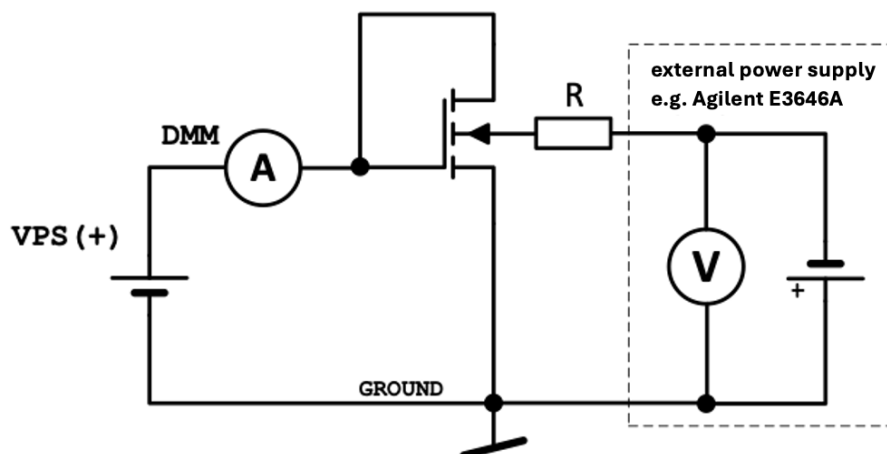


Figure 4: Measurement circuit for determining the body-effect coefficient of the n-MOS transistor.

4.4 Measurement of the Transfer Characteristics of the p-MOS Transistor

Assemble the measurement circuit according to Fig. 5, selecting one of the transistors available in the CD4007 integrated circuit. Using the I-V Plotter, measure a family of transfer characteristics of the p-MOSFET for negative gate voltages U_{GS} ranging from 0 to -10 V, for several values of U_{DS} in the range 0 to -10 V. The U_{GS} step should be approximately 0.2 V (swept automatically by the I-V Plotter), while U_{DS} should be changed in steps of 2 V (set manually on the external power supply), or as instructed by the supervisor.

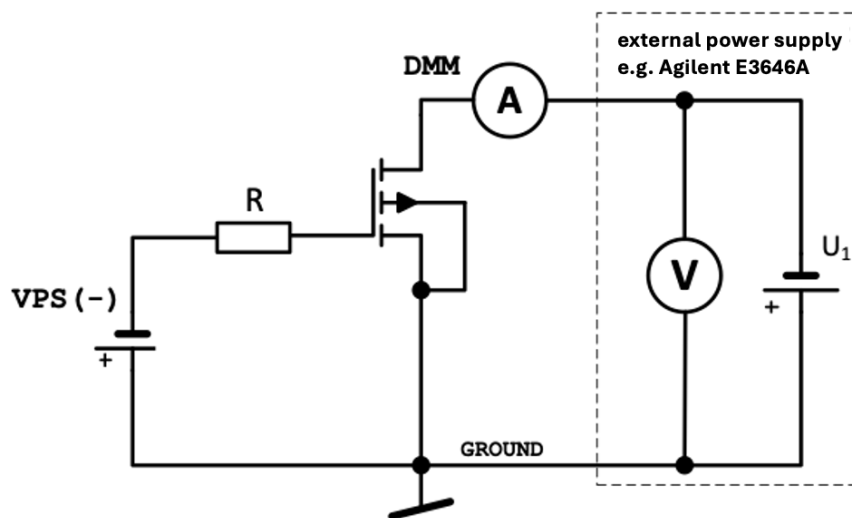


Figure 5: Measurement circuit for the transfer characteristics of the p-MOS transistor.

4.5 Measurement of the Output Characteristics of the p-MOS Transistor

Assemble the measurement circuit according to Fig. 6, selecting one of the transistors available in the CD4007 integrated circuit. Using the I-V Plotter, measure a family of output characteristics of the p-MOSFET for U_{DS} ranging from 0 to -10 V, for several fixed gate voltages (e.g. $U_{GS} = -2, -4, -6, -8, -10$ V), or as instructed by the supervisor.

Note: The I-V Plotter settings may be as follows: start voltage 0 V, stop voltage -10 V, step 0.2 V ($U_{DS \max} = -15$ V). The characteristic will then be plotted in the third quadrant of the coordinate system, starting from 0.

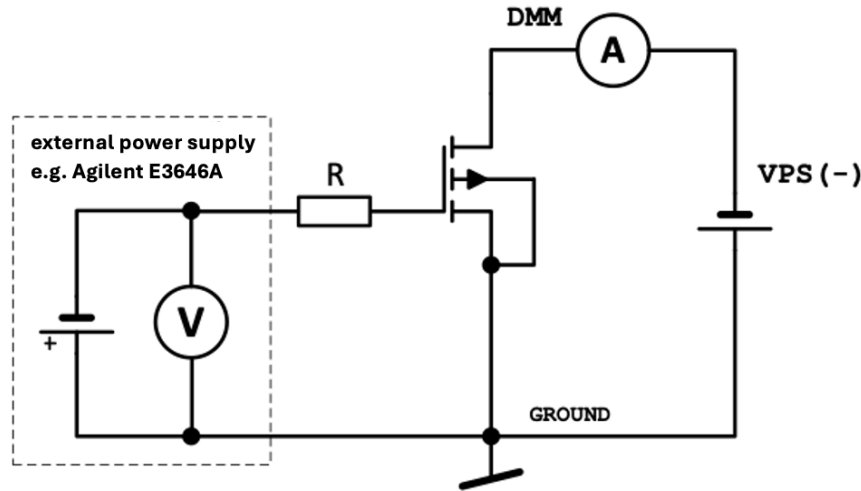


Figure 6: Measurement circuit for the output characteristics of the p-MOS transistor.

4.6 Investigation of the Body Effect in the p-MOS Transistor

The transistor under test should operate in the saturation region. These conditions are ensured by the circuit shown in Fig. 7, when $U_{DS} = U_{GS}$. Assemble the measurement circuit according to Fig. 7, selecting one of the transistors available in the CD4007 integrated circuit.

Using the I-V Plotter, measure a family of transfer characteristics of the p-MOSFET for negative gate voltages U_{GS} ranging from 0 to -10 V, for several fixed values of the bulk-source voltage U_{BS} (e.g. $U_{BS} = 0, 2, 4, 6, 8, 10$ V, step $\Delta U_{GS} = -0.2$ V), or as instructed by the supervisor.

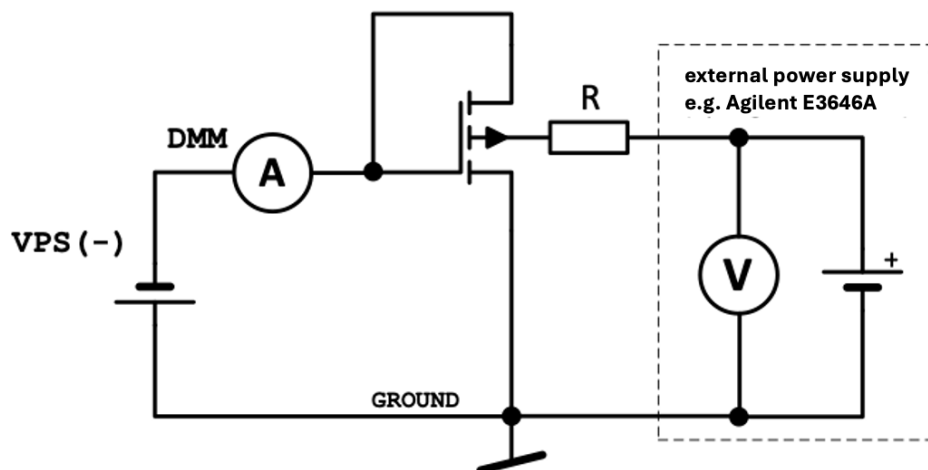


Figure 7: Measurement circuit for determining the body-effect coefficient of the p-MOS transistor.

5 Data Analysis

5.1 n-MOS Transistor

- Using the results collected in Section 4.1, plot the transfer characteristics $I_D = f(U_{GS})$ on a single graph in the lab report. Draw the boundary curve separating the linear (triode) region from the saturation region and label both regions on the graph.
- Plot the auxiliary characteristic $\sqrt{I_D} = f(U_{GS})$ for the transistor operating in saturation and determine the precise value of the threshold voltage V_T .
Note: One curve is sufficient. If the saturation condition ($U_{DS} \geq U_{GS} - V_T$) is not satisfied for all measurement points, restrict the range of those points accordingly.
- Using the results collected in Section 4.2, plot the output characteristics $I_D = f(U_{DS})$ on a single graph in the lab report. Draw the boundary curve separating the linear region from saturation and label both regions. Determine the saturation drain current I_{DSS} for each value of U_{GS} at which output characteristics were measured; present the results in a table. From the above characteristics, determine the transconductance parameter β .
- Using the drain current data collected in Section 4.3, plot the characteristics $\sqrt{I_D} = f(U_{GS})$ for each value of U_{BS} . From these characteristics, determine the body-effect coefficient γ and the transconductance parameter β . Compare the results with those obtained in the previous step and draw conclusions.

5.2 p-MOS Transistor

Using the measurement results obtained in Sections 4.4, 4.5, and 4.6, perform the plots and calculations in an analogous manner to those carried out for the n-MOS transistor.

6 References

- [1] Lecture notes.
- [2] Texas Instruments, *CD4007UB Types: CMOS Dual Complementary Pair Plus Inverter*, Datasheet SCHS051D, Rev. D, 2003. [Online]. Available: <https://www.ti.com/lit/ds/symlink/cd4007ub.pdf>
- [3] onsemi, *BS170: Single N-Channel Small Signal MOSFET, 60 V, 500 mA*, Datasheet BS170/D, Rev. 7, 2023. [Online]. Available: <https://www.onsemi.com/download/data-sheet/pdf/bs170-d.pdf>